Electric field intensity

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Field Due to a Continuous Volume Charge Distribution

- We denote the volume charge density by ρ_v, having the units of coulombs per cubic meter (C/m^3).
- The small amount of charge ΔQ in a small volume Δv is

$$\Delta Q = \rho_v \Delta v$$

- We may define ρ_v mathematically by using a limit on the above equation:

$$\rho_v = \lim_{\Delta v \to 0} \frac{\Delta Q}{\Delta v}$$

- The total charge within some finite volume is obtained by integrating throughout that volume:

$$Q = \int_{\text{vol}} \rho_v dv$$

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Example

Find the total charge inside the volume indicated by $\rho_v = 4xyz^2$, $0 \le \rho \le 2$, $0 \le \Phi \le \pi/2$, $0 \le z \le 3$. All values are in SI units.

$$\left.\begin{aligned} x &= \rho\cos\phi \\ y &= \rho\sin\phi \end{aligned}\right\rangle \rho_v = 4\cdot\rho\sin\phi\cdot\rho\cos\phi\cdot z^2$$

$$Q = \int_{vol} \rho_v dv = \int_{z=0}^{3}\int_{\phi=0}^{\pi/2}\int_{\rho=0}^{2} (4\cdot\rho\sin\phi\cdot\rho\cos\phi\cdot z^2)(d\rho\cdot\rho d\phi\cdot dz)$$

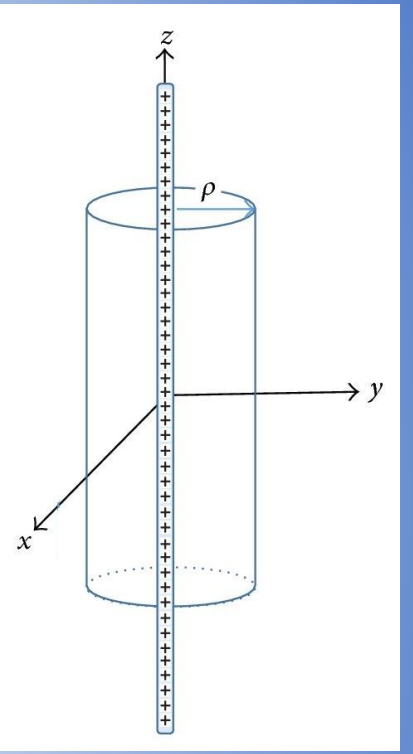
$$= \int_0^3\int_0^{\pi/2}\int_0^2 4\rho^3 z^2 \sin\phi\cos\phi d\rho d\phi dz$$

$$= \int_0^3\int_0^{\pi/2} 16z^2 \sin\phi\cos\phi d\phi dz$$

$$\sin 2\phi = 2\sin\phi\cos\phi$$

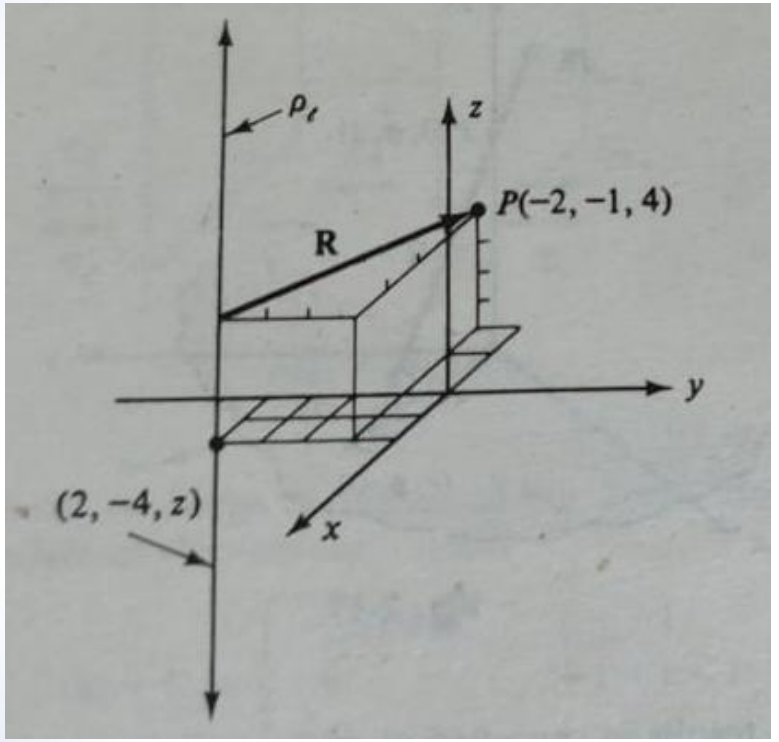
$$= \int_0^3 8z^2 dz = \underline{\underline{72\ \text{C}}}$$

Field of a Line Charge



- Now we consider a filament like distribution of volume charge density. It is convenient to treat the charge as a line charge of density ρ_L C/m.
- Let us assume a straight-line charge extending along the z axis in a cylindrical coordinate system from $-\infty$ to $+\infty$.
- We desire the electric field intensity E at any point resulting from a uniform line charge density ρ_L.
- $$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho$$ (Proof: Not included for exam)
- The field falls off inversely with the distance to the charged line, as compared with the point charge, where the field decreased with the square of the distance.

On the line described by $x = 2\,\text{m}$, $y = -4\,\text{m}$ there is a uniform charge distribution of density $\rho_\ell = 20\,\text{nC/m}$. Determine the electric field $\mathbf{E}$ at $(-2, -1, 4)\,\text{m}$.



$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$

$$\mathbf{R} = -4\mathbf{a}_x + 3\mathbf{a}_y$$

$$\mathbf{E} = \frac{20 \times 10^{-9}}{2\pi\epsilon_0(5)}\left(\frac{-4\mathbf{a}_x + 3\mathbf{a}_y}{5}\right) = -57.6\mathbf{a}_x + 43.2\mathbf{a}_y\ \text{V/m}$$

D2.5. Infinite uniform line charges of 5 nC/m lie along the (positive and negative) x and y axes in free space. Find $\mathbf{E}$ at: (a) $P_A(0, 0, 4)$; (b) $P_B(0, 3, 4)$.

Ans. (a) $45\mathbf{a}_z$ V/m; (b) $10.8\mathbf{a}_y + 36.9\mathbf{a}_z$ V/m


Thank
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